Non-positional Number System

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- Characteristics
  - Each symbol represents the same value regardless of its positions in the number.
  - Symbols are just added to determine a particular number.
  - Example: Roman Number System.

I	1
II	2
V	5
VI	6
X	10

  - Difficulty in Arithmetic operations.

Positional Number System

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- Characteristics
  - Symbols are less, called digits.
  - Digits have different values depending on the position in the number.
  - Example: Decimal Number System.

Positional Number System

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- The value of each digit depends on:
  - Digit itself (face value).
  - Position in the number.
  - Base of the number system.
- Value of a number = $\sum \text{value of the digit} * \text{base}^{\text{index-1}}$

  [index is the position of the digit]
- Base – Total number of different digits in the number system.
  - For any number system maximum value of the of a single digits always equal to one less than the base.

Hexadecimal Number System

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- Positional number system.
- Digits – 0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F.
- Example:

$$\begin{aligned} ABCD_8 &= 13 * 16^0 + 12 * 16^1 + 11 * 16^2 + 10 * 16^3 \\ &= 13 + 192 + 2816 + 40960 \\ &= 43981 \end{aligned}$$

Converting Any Base To Decimal

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- Step 1: Determine the positional value of each digit
- Step 2: Multiply the decimal value of the digit with the positional value.
- Step 3: Summation of these products is the result.

- Value of a number = $\sum decimal\ value\ of\ the\ digit * base^{index-1}$

[index is the position of the digit]

Converting Decimal Value To Any Base

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Division-Remainder method

- Step 1: Divide the decimal value with the base.
- Step 2: Record the remainder from the step 1 as the right most digit (least significant digit).
- Step 3: Go to step 1 with the quotient as the new decimal value until the quotient is 0.

Fractional Number

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Decimal to any base

- Step 1: Multiply the fractional number with the Base.
- Step 2: Record the non-fractional part of the resulting number from leftmost digit after fractional point.
- Step 3: Go to step 1 with the fractional part of the resulting number (stop if the fractional part is 0 or if significant number of digits are calculated).

Fractional Number

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Convert 0.865 from Base 10 to Base 2,8 and 16.

0.865	0.865	0.865
x 2	x 8	x 16
1.730	6.920	5190
x 2	x 8	865 x
1.460	7.360	13.840
x 2	x 8	x 16
0.920	2.880	5040
x 2	x 8	840 x
1.840	7.040	13.440
x 2		x 16
1.680	The octal equivalent of $(0.865)_{10}$ is $(.6727)_8$	2640
x 2		440 x
1.360		7.040
The binary equivalent of $(.865)_{10}$ is $(.110111)_2$		The number 13 in hexadecimal is D. The hexadecimal equivalent of $(0.865)_{10}$ is $(.DD7)_{16}$

Conversion among two bases

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- Step 1: Convert the number from "From Base" to base 10.
- Step 2: Convert the number in base 10 to "To Base" number.

Shortcut method (Hexadecimal to Binary)

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$2AB_{16} = ?_2$

Step 1: Convert each hexadecimal digit to a 4 digit binary number

$2_{16} = 2_{10} = 0010_2$
$A_{16} = 10_{10} = 1010_2$
$B_{16} = 11_{10} = 1011_2$

Step 2: Combine the binary groups

$2AB_{16} = \underset{2}{0010}\ \underset{A}{1010}\ \underset{B}{1011}$

Hence, $2AB_{16} = 001010101011_2$

Equivalent Chart

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Decimal	Binary	Octal	Hexadecimal
0	0000	000	0
1	0001	001	1
2	0010	002	2
3	0011	003	3
4	0100	004	4
5	0101	005	5
6	0110	006	6
7	0111	007	7
8	1000	010	8
9	1001	011	9
10	1010	012	A
11	1011	013	B
12	1100	014	C
13	1101	015	D
14	1110	016	E
15	1111	017	F
16	10000	020	10